

# Low Complexity Early Termination HARQ for URLLC: Analysis and Neural Network Design

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**Abstract**—This paper presents the analysis and a proof-of-concept design of the low-complexity early termination hybrid automatic repeat request (ET-HARQ) for ultra-reliable low-latency communication (URLLC). In ET-HARQ, unfit packets are flagged for retransmission prior to decoding, a process that significantly diminishes decoding complexity and facilitates swift HARQ reporting. This characteristic makes ET-HARQ well-suited for URLLC applications. We analyze the impact of ET-HARQ on the packet error rate (PER), throughput, and complexity performance in the finite block length regime, taking into consideration cyclic redundancy check (CRC) limitations. Numerical results indicate that ET-HARQ significantly reduces the decoding complexity and improves throughput with little to no loss in the PER. In addition, ET-HARQ demonstrates resilience even with a short CRC, whereas the imperfections of a short CRC significantly impact PER reliability in standard HARQ. To validate our analysis, we also design a practical early termination mechanism involving belief propagation and neural network (BP-NN) to predict the decodability of the received packet. Testing with BCH and CRC-polar codes shows that it can reach up to a 70 % ~ 80 % prediction accuracy with packet lengths less than 128 bits encoded with high-density linear block codes. Simulation shows that the BP-NN-based ET-HARQ has significantly lower complexity at this accuracy level than the standard HARQ with similar reliability.

**Index Terms**—CRC, early decoding, fast HARQ, finite block length, low complexity decoding, ordered statistic decode, URLLC.

## I. INTRODUCTION

THE main focus of previous generations of mobile communication systems was to enable enhanced mobile broadband services for the purpose of delivering large files with high reliability and data rate [1]. Therefore, the communication design did not prioritize the latency of individual

data packets. As a result, long packets of data with multiple retransmission attempts of failing packets and complex encoder-decoder were successfully incorporated into the communication design to improve average packet error rate (PER) and throughput performance. The fifth generation (5G) and the future sixth generation (6G) of mobile standards are envisioned to support new services classified as Ultra-reliable low latency communication (URLLC) and massive machine type communication (mMTC) [2]. URLLC and mMTC are key enablers of services that can revolutionize many critical sectors of society, such as smart cities, smart agriculture, telehealth, industrial Internet of things for smart manufacturing, etc. [3]. Most of these services require communication with a target to achieve user plane latency below 1 ms and reliability above 99.9999% for a packet of size 32 bytes with or without retransmission [4].

To meet such strict latency-reliability targets, the research on the design of powerful short block-length codes and efficient decoding techniques is prioritized [5], [6]. In the finite blocklength (FBL) regime, a performance gap to the Shannon limit is inevitable [7]. To address this, URLLC systems often employ mini-slot durations, allowing for one or two retransmissions of failing packets within the stringent packet-level latency budget [8]. While practical near maximum-likelihood (ML) decoding algorithms, such as order statistics decoding (OSD), can achieve performance close to the normal approximation bound [9] in the FBL regime, they come with significantly high decoding complexity [10]. Higher decoding complexity not only increases the latency but also limits the application's scope, as many URLLC applications involve low-cost receivers with limited power. Thus, reliability, latency, and decoding complexity are intrinsically linked and equally critical for the success of URLLC [11]. For instance, while retransmissions enhance reliability by leveraging time and frequency diversity, they also introduce additional decoding effort per information block, which can increase latency. Therefore, a low-complexity decoding mechanism combined with a limited number of retransmission rounds is highly desirable for URLLC [12].

## A. Related Work

Packet retransmission schemes such as automatic repeat request (ARQ) can enhance packet reliability by providing additional opportunities for lost packets and introducing time and frequency diversity, making the communication more resilient to channel fading and interference [13], [14]. Furthermore, retransmissions combined with error correction

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known as hybrid-ARQ (HARQ) gradually increase the effective packet length and improve the coding gain [15]. This ensures efficient resource utilization by dynamically adapting the coding rate to channel conditions, with minimum overhead [16]. Due to these reasons, HARQ has been the backbone of contemporary communications standards such as fourth-generation (4G)

In standard HARQ, multiple retransmissions are allowed using feedback, where each round involves full decoding followed by a cyclic redundancy check (CRC) to identify the success or failing states of the packets. The full decoding of the failing packets in multiple attempts is a significant computational overhead that can cause additional delays in upcoming packets [17]. Moreover, additional CRC-bits cause significant throughput loss due to short block length, and fewer CRC-bits can cause unnecessary retransmission due to poor error detection capability [18]. These limitations restrict the use of standard HARQ retransmission for URLLC applications.

Significant effort has been dedicated to designing an effective HARQ scheme to facilitate URLLC applications [13], [14], [17], [19], [20]. In the standard incremental redundancy HARQ (IR-HARQ) method, the transmitter sends the encoded packet with more parity bits (redundancy) whenever the decoding fails. With each retransmission attempt of IR-HARQ, the redundancy gradually increases, which increases the packet length and coding gain [21]. It is often assumed that the decoding process under HARQ happens instantly without any decoding delay. This assumption is not valid and cannot be ignored in the case of URLLC [13], [17], [19]. Furthermore, a high percentage of packet retransmissions, with a full, decoding attempt, after each retransmission round, can substantially increase the overall decoding complexity and delay [11]. Also, the additional decoding delay of HARQ during retransmission can cause queuing delays for the upcoming packets, which is highly deleterious for URLLC services.

To reduce decoding complexity for both single transmission and HARQ systems, there has been ongoing effort to develop early termination techniques within decoders [22], [23]. For example, in belief propagation (BP) decoding, the decoder can avoid decoding the non-convergent log-likelihood ratios (LLRs) by monitoring their mean magnitude stability [24]. Similarly, in successive cancellation (SC) decoding, using CRC checks, the decoding can be terminated earlier. Although these early termination inside decoder reduces decoding complexity to some extent, it still demands sophisticated termination criteria, and decoding all received packets, including those that eventually fail, results in substantial computation overhead. In HARQ, since failing packets are retransmitted, attempting to decode every packet, even with a low complexity decoder, can be inefficient.

To address this, we propose an early termination HARQ (ET-HARQ) scheme where the early termination is applied before decoding. It involves a strategy where packets do not go to decoding process immediately upon arrival. Instead, preliminary quality tests are conducted to assess the likelihood of successful or failed decoding. These tests may utilize metrics like signal-to-noise ratio (SNR), log-likelihood ratios, or channel quality or more sophisticated techniques such as preamble-based detection or subcode-based partial decoding,

etc. [25], [26]. Based on these tests, packets deemed unlikely to be decoded successfully are terminated early, and the receiver waits for their retransmission while only promising packets are selected for decoding [27]. This distinction highlights a fundamental shift from traditional decoder-specific ET methods toward a pre-decoding strategy that is agnostic to the underlying decoding algorithm. Additionally, our ET mechanism can complement existing decoder-specific ET methods.

The proposed ET mechanism is designed using a computationally efficient NN. Recently, very advanced procedures like NN-based classification of decodable and undecodable packets have become more popular for their performance. This is because NN can exploit various quality metrics and code structures to improve prediction accuracy [28], [29]. While NNs have also been explored for complete channel decoding [30], [31], such decoders typically require massive architectures (100K-10M parameters) because they must solve the NP-hard problem of identifying the correct codeword from an exponential-size codebook. Moreover, recent theoretical analysis [32] shows that achieving ultra-low error rates with learned decoders requires prohibitively large training sets. This motivates using NNs solely for pre-decoding ET classification (decodable vs. undecodable), a simpler binary task that keeps the network dimension minimal.

## B. Contributions

In this paper, we study the performance of ET-HARQ in terms of PER, decoding complexity, and throughput and design an effective ET-HARQ approach. Specifically,

- We propose ET-HARQ to highlight the nature of the complexity-reliability trade-off with early termination. In ET-HARQ, the predicted undecodable packet (using the early termination technique) can be dropped without performing full decoding, leading to substantial complexity reduction. However, since ET-HARQ is applied in all rounds, this can lead to reliability loss. To overcome this, we use selective ET-HARQ, where early termination (prediction) is used only for calling retransmission, whereas in the last transmission round, all packets undergo full decoding to avoid any loss of decodable packets. The selective ET-HARQ also shows significant complexity reduction without losing PER reliability.
- We analyze the PER, throughput, and complexity for the proposed ET-HARQ scheme, starting with a two-round transmission case and extending it to the general scenario. Numerical simulations are conducted to evaluate the performance of the proposed ET-HARQ schemes with low and moderate prediction accuracies. We also highlight the positive impact of ET-HARQ when considering realistic CRC lengths for short packet communication, which is different from most previous studies on HARQ that assume ideal CRC and ignore its impact on reliability. Our results indicate that the early terminations mechanism can 1) significantly reduce the overall decoding complexity at the receiver and 2) mitigate the reliability loss introduced by the undetected errors of short CRC.
- We develop a proof-of-concept early termination mechanism using BP and NN to showcase the feasibility and

potential of ET-HARQ. Upon receiving a packet from the channel, the BP-NN-based early termination first refines the symbol LLRs using a low-complexity BP decoder. Then, a trained NN evaluates whether the received block is decodable by a high complexity, near ML decoder. Simulation results with Bose–Chaudhuri–Hocquenghem (BCH) codes and CRC-polar codes demonstrate that the proposed BP-NN-based ET-HARQ significantly reduces decoding complexity and increases throughput, aligning with our analysis.

The rest of the paper is organized as follows: Section II introduces the system model and preliminaries. In Section III, we present the reliability and complexity analysis of standard HARQ for performance benchmark. We propose ET-HARQ and analyze its performance in Section IV. Section V presents numerical results for the proposed ET-HARQ scheme in various scenarios. In Section VI, we develop a BP-NN-based early termination technique, illustrating the practical potential of ET-HARQ schemes. Finally, Section VII concludes the paper.

## II. SYSTEM MODEL AND PRELIMINARIES

### A. Standard HARQ

We consider a point-to-point discrete-time communication system in a binary-input additive white Gaussian noise (BI-AWGN) channel with SNR  $\gamma$ . We use short block length codes with IR-HARQ to send  $k$  information bits in maximum  $M$  transmission rounds. More specifically, a binary information sequence  $\mathbf{u} = [u_1, u_2, \dots, u_k]$ ,  $u_i \in \{0, 1\}$ , of length  $k$  is encoded to generate a codeword  $\mathbf{c}_1 = [c_1, c_2, \dots, c_{n_1}]$  of length  $n_1$ . We use the binary phase shift keying (BPSK) modulation for bits-to-symbol mapping and generate a length- $n_1$  sequence of BPSK symbols,  $\mathbf{x}_1 = [x_1, x_2, \dots, x_{n_1}]$ , such that  $x_i = (-1)^{c_i}$ . Upon failed decoding, the receiver feedbacks negative acknowledgment (NACK), and the transmitter sends  $n_2 - n_1$  additional coded symbols to the receiver to increase the effective blocklength to  $n_2$  and accordingly reduce the code rate to  $R_2 = k/n_2$ . The transmitter keeps sending additional coded symbols to the receiver, i.e.,  $n_m - n_{m-1}$  coded symbols in the  $m$ -th transmission round, for  $1 \leq m \leq M$ , upon receiving every NACK from the receiver, until all  $M$  transmission rounds are completed or the receiver successfully decodes the message and sends an acknowledgment (ACK) to the transmitter. Upon completing round  $m$  of the transmission, the receiver attempts decoding the codeword  $\mathbf{c}_m = [c_1, c_2, \dots, c_{n_m}]$  of length  $n_m$  and rate  $R_m = k/n_m$ . To generate coded symbols, we consider puncturing of a mother code of rate  $R_M = k/n_M$  with generator matrix  $\mathbf{G}^{(M)}$  to obtain  $\mathbf{G}^{(m)}$ , which denotes the generator matrix corresponding to the codeword  $\mathbf{c}_m$  of rate  $R_m$ , i.e.,  $\mathbf{c}_m = \mathbf{u}\mathbf{G}^{(m)}$ .

### B. Packet Error Rate at FBL

The packet length ( $n_m$ ) increases with each retransmission round, and the receiver gets another chance to decode the packet with a higher error correction ability. However, short block-length packet transmission incurs performance loss compared to the Shannon capacity limit due to the channel dispersion and limited channel observations [7]. According to the normal approximation [7], the achievable rate at given

codeword length  $n_m$ , SNR  $\gamma$ , and PER  $\epsilon_m$ , is approximated by

$$R(n_m, \gamma, \epsilon_m) = C(\gamma) - \sqrt{\frac{V(\gamma)}{n_m}} Q^{-1}(\epsilon_m) \log e + \mathcal{O}\left(\frac{\log n_m}{n_m}\right), \quad (1)$$

where  $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{v^2}{2}} dv$  is the standard Q-function, and for a BI-AWGN channel at SNR  $\gamma$ , we have  $C(\gamma) = 1 + \frac{H^{(1)}(0)}{\ln(2)}$ , and  $V(\gamma) = H^{(2)}(0) - (H^{(1)}(0))^2$ , where

$$H^{(\ell)}(0) = \frac{1}{\sqrt{2\pi\gamma}} \int_{-\infty}^{\infty} e^{-\frac{(x-\gamma)^2}{2\gamma}} (-h(x))^\ell dx, \quad (2)$$

and  $h(x) = \ln(1 + e^{-2x})$  [9].

Accordingly, the PER can be approximated at the  $m$ -th round of the transmission as [9]

$$\epsilon_m(n_m, k, \gamma) \approx Q\left(\frac{n_m C(\gamma) - k + \log_2(n_m)}{\sqrt{n_m V(\gamma)}}\right). \quad (3)$$

In the rest of the paper, we use  $\epsilon_m$  instead of  $\epsilon_m(n_m, k, \gamma)$  for notational simplicity.

Following [19], [20], we further assume that it is almost impossible that a packet could have been decoded after  $m - 1$  transmission rounds when the receiver failed to decode it after  $m$  rounds of transmission. This assumption is formally described as follows.

*Assumption 1:* Let  $\mathcal{E}_m$  denote the event that the decoding fails at the receiver after  $m$  rounds of transmissions, for  $m > 1$ , where  $\text{Prob}\{\mathcal{E}_m\} = \epsilon_m$ . Let  $\epsilon'_m = \text{Prob}\{\mathcal{E}_m | \mathcal{E}_{m-1}\}$ , which denotes the probability that the decoding fails after  $m$  rounds of transmission, given that the decoding has failed after  $m - 1$  rounds of transmissions. We assume that  $\text{Prob}\{\mathcal{E}_{m-1} | \mathcal{E}_m\} \approx 1$ ; indicating that if a packet failed after  $m$  transmission rounds, it almost certainly failed with  $m - 1$  transmission rounds. Using this assumption, we can easily show that

$$\epsilon_{m-1} \epsilon'_m \approx \epsilon_m. \quad (4)$$

In fact  $\epsilon_{m-1} \epsilon'_m = \text{Prob}\{\mathcal{E}_{m-1}\} \text{Prob}\{\mathcal{E}_m | \mathcal{E}_{m-1}\} = \text{Prob}\{\mathcal{E}_m \cap \mathcal{E}_{m-1}\}$ . Moreover,  $\text{Prob}\{\mathcal{E}_m \cap \mathcal{E}_{m-1}\} = \text{Prob}\{\mathcal{E}_m\} \text{Prob}\{\mathcal{E}_{m-1} | \mathcal{E}_m\} \approx \text{Prob}\{\mathcal{E}_m\} = \epsilon_m$ , which stems from the assumption  $\text{Prob}\{\mathcal{E}_{m-1} | \mathcal{E}_m\} \approx 1$ .

The assumption in (4) clarifies the process of calculating PER and makes the subsequent analysis easier to follow.

### C. Decoding Complexity at FBL

The PER calculated in (3) is based on the normal approximation [7] and can be achieved by employing the best short codes paired with a (near) ML decoding [5]. As reported in [5] and [12], such code-decoder pairs include BCH codes OSD and CRC-polar codes with SC list/stack decoding. Any decoding algorithms can be treated as the process through which information is cleaned from noise, and there exists a trade-off between decoding complexity and the PER. A received codeword that is less impacted by noise can be correctly decoded with less decoding complexity. Particularly, the



undecodable blocks will make the decoder achieve the worst-case complexity<sup>1</sup> because decoder must explore all potential decoding solutions [12]. Conversely, decodable blocks lead to significantly reduced decoding complexity, as decoders can terminate early upon detecting the correct output [12].

For analysis purposes, we assume near ML decoding and use the general rules to assign decoding complexity without going into the exact decoding method. Let  $K_s^{(m)}$  and  $K_f^{(m)}$  denote the average number of operations in decoding a successful (decodable) or a failing (undecodable) packet of length  $n_m$ , respectively. Then, we have  $K_s^{(m)} \leq K_f^{(m)}$ . This is because  $K_f^{(m)}$  is the worst-case complexity of the decoder, while  $K_s^{(m)}$  usually represents the average decoding complexity excluding decoding attempts of undecodable blocks. Furthermore, the decoding complexity also varies with respect to transmission rate  $R_m = k/n_m$ , as  $K_s^{(m-1)} \leq K_s^{(m)}$  and  $K_f^{(m-1)} \leq K_f^{(m)}$ , which stems from the fact that decoding lower rate codes require more operations to achieve near ML performance. Note that  $K_s^{(m)}$  and  $K_f^{(m)}$  can be derived from simulations or theoretical analysis for a specific decoder type and typical rate  $R_m = k/n_m$ . In Section V, we elaborate this further with exact complexity expression for the OSD decoder.

#### D. Impact of Decoding Complexity on Latency

In URLLC applications, decoding latency dominates the total receiving latency [1], [33], while other delay components remain relatively minor. The transmission delay, which includes encoding, channel propagation, and symbol transmission time, is proportional to the blocklength and thus remains small for short packets ( $n < 200$ ). Similarly, the processing delays for operations such as demodulation, channel estimation, and synchronization are negligible due to their linear complexity and hardware-efficient implementations [34]. However, the decoding delay presents a significant challenge due to the computational burden of probabilistic or algebraic decoding algorithms. These algorithms, designed to approach ML performance in URLLC, require operations that far exceed the computational requirements of other system components. For instance, the near-ML OSD decoder employed in this work exhibits complexity that grows exponentially with the decoding order, as detailed in Appendix. This exponential complexity leads to substantial decoding delays that persist even with longer codes. The challenge is particularly acute in HARQ systems, where multiple decoding attempts may be required for each packet. While many HARQ studies [17], [19] assume instantaneous retransmissions and ignore decoding delays, practical implementations must account for this significant latency component. Our proposed ET mechanism specifically addresses this limitation by selectively avoiding unnecessary decoding attempts, thereby reducing the overall impact of decoding delay in retransmission systems.

### III. ANALYSIS OF CRC-ENABLED STANDARD HARQ

The CRC-enabled standard HARQ, as summarized in Fig. 1, governs packet transmission and retransmission. The receiver

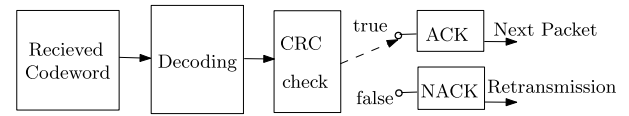


Fig. 1. CRC-enabled standard HARQ.

performs full decoding of each received codeword, followed by a CRC check to identify whether the decoding is erroneous or not. If no errors are detected in the decoded packet, indicated by the CRC check being true in Fig. 1, the receiver sends an ACK, and the transmitter proceeds to the next packet. On the other hand, when the CRC check is false, a NACK is fed back to request further retransmission. After each retransmission, the receiver decodes the codeword with increased packet length due to additional parity being released. Finally, if the receiver is unable to recover a failing packet while the maximum number of transmission rounds, i.e.,  $M$  is reached, the packet is dropped.

The CRC check involves appending a fixed-length checksum to the data, calculated using a polynomial division algorithm. Upon reception, the receiver recalculates the CRC and compares it with the received CRC; any mismatch indicates the presence of errors in the data. A  $k_C$  bits CRC encoding has a probability that an erroneous codeword goes undetected as  $p_{ud} = 2^{-k_C}$  [35]. In the study of HARQ, a perfect CRC, capable of accurately identifying the true packet decoding state, is often assumed for simplicity [20], [21]. This assumption is valid only when the packet length is much larger than CRC bits to accommodate more CRC bits. However, it is essential to recognize that in short packet communication increasing CRC-bits to improve reliability can cost significant throughput overhead. A short CRC has a higher likelihood of undetected failed packets leading to packet errors due to missed retransmission opportunities.

#### A. Special Case With $M = 2$ Transmissions

Next, in the following three Lemmas, we derive the PER reliability, throughput, and decoding complexity of HARQ under non-ideal CRC when a single retransmission is allowed, i.e.,  $M = 2$ .

*Lemma 1:* The PER of CRC-enabled standard HARQ, denoted by  $\zeta_S$ , when  $M = 2$  is given by:

$$\zeta_S \approx p_{ud}\epsilon_1 + (1 - p_{ud})\epsilon_2, \quad (5)$$

where  $\epsilon_m$  is the error probability in the  $m$ -th transmission round defined in (3).

*Proof:* Since  $M = 2$ , there are a maximum of two rounds of transmission for each packet, i.e.,  $m \in \{1, 2\}$ . The packet error probability with single transmission is  $\epsilon_1$ . However, due to an imperfect CRC,  $p_{ud}\epsilon_1$  erroneous packets remain undetected and are dropped from the retransmission opportunity. Therefore, retransmission is only requested for  $(1 - p_{ud})\epsilon_1$  packets, detected as erroneous when  $m = 1$ . With the second round of transmission, the packet length increases to  $n_2$ , and decoding is performed again to recover the failing packets. Among these retransmitted packets,  $(1 - p_{ud})\epsilon_1(1 - \epsilon'_2)$  packets succeed, while  $(1 - p_{ud})\epsilon_1\epsilon'_2$  fail, where  $\epsilon'_2$  is the PER at length  $n_2$  given that a packet with length  $n_1$  has failed. Then,

<sup>1</sup>In practice, this worst-case complexity is usually restricted by the maximum number of allowed operations (e.g., maximum visited tree nodes of SC and the maximum decoding order of the OSD).

according to the assumption (4), i.e.,  $\epsilon_1 \epsilon'_2 \approx \epsilon'_2$ , the PER after retransmission is approximated by  $(1 - p_{\text{ud}})\epsilon_2$ . The overall PER is then  $p_{\text{ud}}\epsilon_1 + (1 - p_{\text{ud}})\epsilon_2$ , which completes the proof. ■

**Lemma 2:** The average decoding complexity of the CRC-enabled standard HARQ, when  $M = 2$ , is

$$K_S \approx (1 - \epsilon_1)K_s^{(1)} + \epsilon_1 K_f^{(1)} + (1 - p_{\text{ud}}) \left( (\epsilon_1 - \epsilon_2)K_s^{(2)} + \epsilon_2 K_f^{(2)} \right). \quad (6)$$

*Proof:* In the first round of transmission, the packet is decoded successfully with probability  $1 - \epsilon_1$  and complexity  $K_s^{(1)}$ , and failed the decoding with probability  $\epsilon_1$  and complexity  $K_f^{(1)}$ . Therefore, the average decoding complexity in the first round of transmission is  $(1 - \epsilon_1)K_s^{(1)} + \epsilon_1 K_f^{(1)}$ . A failed packet in the first round of transmission will be detected through the CRC check with probability  $(1 - p_{\text{ud}})$ , and the decoding will be repeated after the second transmission round. This packet will be decoded with probability  $1 - \epsilon'_2$  and complexity  $K_s^{(2)}$  and failed the decoding with probability  $\epsilon'_2$  and complexity  $K_f^{(2)}$ . Therefore, the average complexity in the second round of transmission is  $(1 - p_{\text{ud}})\epsilon_1 \left( (1 - \epsilon'_2)K_s^{(2)} + \epsilon'_2 K_f^{(2)} \right)$ , which can be further simplified using the assumption in (4) to  $(1 - p_{\text{ud}}) \left( (\epsilon_1 - \epsilon_2)K_s^{(2)} + \epsilon_2 K_f^{(2)} \right)$ . The overall average decoding complexity is then the sum of average decoding complexities in the first and second rounds of transmission. This complete the proof. ■

**Lemma 3:** The throughput of CRC-enabled standard HARQ, with  $k_C$ -bit CRC, when  $M = 2$ , is

$$\eta_S \approx k_I \frac{1 - \zeta_S}{K_S T_b}, \quad (7)$$

where  $k_I = k - k_C$  and  $T_b$  is the time required for a binary operation on the decoder hardware platform.

*Proof:* This can be directly proved using Lemma 1 and Lemma 2, and based on the definition of throughput, which is the ratio of the average number of bits that have been successfully delivered to the receiver and the total time required for decoding them at the receiver. ■

### B. General Case With $M$ Transmissions

The analysis presented for two-round transmission can be extended to the general case with  $M$  transmission rounds. As illustrated in Fig. 1, the receiver follows identical decision steps after each transmission to either request retransmission or proceed to a new packet. This process continues recursively until either successful decoding or exhausting all  $M$  transmissions.

Let  $p_m^t$  denote the probability that transmission stops after the  $m$ -th round (due to successful decoding or undetected error), and  $p_m^r$  denote the probability of requesting the  $(m+1)$ -th transmission (due to NACK), where  $1 < m < M$ . The recursive relationships for these probabilities are:

$$p_m^t \approx p_{m-1}^t (1 - \epsilon_m) + p_{m-1}^r p_{\text{ud}} \epsilon_m, \quad (8)$$

$$p_m^r \approx p_{m-1}^r (1 - p_{\text{ud}}) \epsilon_m. \quad (9)$$

Here, (8) accounts for both successful decoding  $(1 - \epsilon_m)$  and undetected errors  $(p_{\text{ud}} \epsilon_m)$ , while (9) represents detected errors leading to retransmission. For  $m = 1$ , these equations reduce to  $p_1^t \approx (1 - \epsilon_1) + p_{\text{ud}} \epsilon_1$  and  $p_1^r \approx (1 - p_{\text{ud}}) \epsilon_1$ , consistent with the case described in (5).

Using these recursive relationships, we can determine all  $p_m^t$  and  $p_m^r$  values. The packet error occurs when errors remain undetected at any transmission round or when the final transmission fails. Thus, the PER for a general  $M$ -round HARQ is:

$$\zeta_S(M) \approx p_{\text{ud}} \epsilon_1 + \sum_{m=2}^{M-1} p_{m-1}^r p_{\text{ud}} \epsilon_m + p_{M-1}^t \epsilon_M, \quad (10)$$

where the first term represents undetected errors in the initial transmission, the summation accounts for intermediate rounds, and the final term captures errors in the last transmission (which does not use CRC). Setting  $M = 2$  in (10) retrieves (5).

Following similar reasoning, we derive the average complexity by considering decoding attempts at each transmission round. The first round always requires decoding with complexity  $K_s^{(1)}$  for success  $(1 - \epsilon_1)$  or  $K_f^{(1)}$  for failure  $(\epsilon_1)$ . For subsequent rounds  $m > 1$ , decoding occurs with probability  $p_{m-1}^r$  (the probability of reaching round  $m$ ), and requires complexity  $K_s^{(m)}$  for successful decoding  $(1 - \epsilon_m)$  or  $K_f^{(m)}$  for failed attempts  $(\epsilon_m)$ . Thus, the average complexity for  $M$ -round transmission is:

$$K_S(M) \approx K_s^{(1)}(1 - \epsilon_1) + K_f^{(1)} \epsilon_1 + \sum_{m=2}^M p_{m-1}^r \left( K_s^{(m)}(1 - \epsilon_m) + K_f^{(m)} \epsilon_m \right). \quad (11)$$

Finally, the throughput can be obtained as

$$\eta_S(M) \approx k_I \frac{1 - \zeta_S(M)}{K_S(M) T_b}, \quad (12)$$

which is obtained similarly to (7).

**Remark 1:** Note that our analysis framework accounts for various CRC failure scenarios, including those encountered in decoder-specific ET methods. A key component of our framework is the parameter  $p_{\text{ud}}$  which models the probability of undetected errors in any CRC-based error detection scheme. This parameter enables a unified analysis that is not tied to any specific decoding algorithm using CRC, making it applicable to standard HARQ, CRC-Polar codes, and other CRC-based methods [24], [36]. For example, in HARQ processes where CRCs are used for error detection, there is always a nonzero probability of undetected errors that may lead to premature retransmission termination. Our analysis captures this behavior and explicitly models the implications of such errors on system performance. This generality ensures that our proposed ET mechanism accounts for all potential CRC-based error scenarios, providing accurate predictions of throughput under realistic conditions.

## IV. EARLY TERMINATION HARQ WITH CRC

Fig. 2 present the proposed ET-HARQ scheme. All the received codewords go through an early termination (ET)

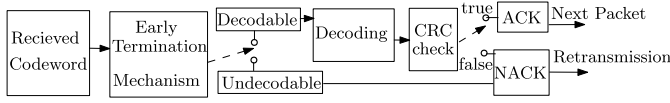


Fig. 2. CRC-enabled Early Termination HARQ.

mechanism. The ET mechanism utilizes some preliminary tests to predict whether the received codeword is decodable or undecodable. Based on the prediction outcome, if the packet is found to be decodable, the receiver performs full decoding; otherwise, it directly requests a retransmission through NACK. The packet with full decoding undergoes the usual retransmission via ACK or NACK based on the CRC check being true or false, respectively.

Without loss of generality, we assume that the ET mechanism in the  $m$ -th transmission round detects a decodable packet correctly with probability  $\alpha_m$  and falsely declares a decodable packet as undecodable, with probability  $1 - \alpha_m$ . We also assume that the early termination mechanism correctly detects the undecodable packet in the  $m$ th transmission round with probability  $\beta_m$ , and falsely declares an undecodable packet decodable with probability  $1 - \beta_m$ . The confusion matrix for the early termination mechanism in the  $m$ th round of transmission is depicted in Fig. 3.

#### A. Performance Analysis of ET-HARQ With CRC When $M = 2$

The PER of the ET-HARQ with CRC is given in the following Lemma

**Lemma 4:** The PER of ET-HARQ with CRC, denoted by  $\zeta_E$ , when  $M = 2$  is given by:

$$\begin{aligned} \zeta_E \approx & (1 - \epsilon_1)(1 - \alpha_1)(1 - \alpha_2) + \epsilon_1(1 - \beta_1)p_{ud} \\ & + ((1 - \beta_1)(1 - p_{ud}) + \beta_1)((\epsilon_1 - \epsilon_2)(1 - \alpha_2) + \epsilon_2). \end{aligned} \quad (13)$$

*Proof:* In the ET-HARQ, the PER is determined by considering events where packets are either declared failed after decoding with CRC or dropped due to ET mechanism, preventing further retransmission or decoding. A key event occurs when the receiver prematurely terminates decoding of a potentially decodable packet due to false prediction. In particular, the probability that a packet is decodable in the first round of transmission is  $1 - \epsilon_1$ , which is declared undecodable by the ET mechanism with probability  $1 - \alpha_1$ . After the second round of transmission, this packet is again declared undecodable with probability  $1 - \alpha_2$  by the ET mechanism. Therefore, the probability that a packet is decodable in the first round of transmission but is declared undecodable in both rounds of transmissions is  $q_1 = (1 - \epsilon_1)(1 - \alpha_1)(1 - \alpha_2)$ .

A packet in the first round of transmission is undecodable with probability  $\epsilon_1$ , which can be falsely declared as decodable with probability  $1 - \beta_1$ . After the decoding, the probability that the CRC check falsely declares the decoding successful is  $p_{ud}$ . Therefore, the probability that in the first round of transmission, an undecodable packet goes through decoding and CRC bits are falsely checked is  $q_2 = \epsilon_1(1 - \beta_1)p_{ud}$ .

The probability that a packet is undecodable in the first round of transmission is  $\epsilon_1$ , which is correctly declared as

|                       | Actual Decodable | Actual Undecodable |
|-----------------------|------------------|--------------------|
| Predicted Decodable   | $\alpha_m$       | $1 - \beta_m$      |
| Predicted Undecodable | $1 - \alpha_m$   | $\beta_m$          |

Fig. 3. Confusion matrix of the ET mechanism at the  $m$ -th transmission round.

undecodable with probability  $\beta_1$ . This packet is also undecodable in the 2nd round of transmission with probability  $\epsilon'_2$ . Therefore, the probability that a packet is undecodable and correctly declared undecodable in the first round of transmission and remained undecodable in the second round of transmission is  $q_3 = \epsilon_1\beta_1\epsilon'_2 \approx \beta_1\epsilon_2$ , where the approximation is due to (4).

A packet is undecodable in the first round of transmission with probability  $\epsilon_1$  and will undergo retransmission, if it is declared undecodable with ET (which happens with probability  $\beta_1$ ), or if it is falsely declared decodable (with probability  $1 - \beta_1$ ) but after the decoding, the CRC correctly detect the false decoding (which occurs with probability  $(1 - \beta_1)(1 - p_{ud})$ ). Therefore, the probability that a packet is undecodable in the first round of transmission and will undergo retransmission in the second round is  $\epsilon_1((1 - \beta_1)(1 - p_{ud}) + \beta_1)$ . Such a packet is decodable in the second round of transmission with probability  $1 - \epsilon'_2$ , and is undecodable with probability  $\epsilon'_2$ . If it is decodable, it is falsely declared undecodable in the second round of transmission, with probability  $1 - \alpha_2$ . Therefore, the probability that a packet which is undergone the second transmission round is deemed undecodable is  $(1 - \epsilon'_2)(1 - \alpha_2) + \epsilon'_2$ . Therefore, the probability that a packet is undecodable in the first round of transmission, is undergone the second round of transmission, and is deemed undecodable either through the ET mechanism or the decoding algorithm and CRC check, is  $q_4 = \epsilon_1((1 - \beta_1)(1 - p_{ud}) + \beta_1)((1 - \epsilon'_2)(1 - \alpha_2) + \epsilon'_2)$ . By using approximation (4) and summing  $q_1, q_2, q_3$ , and  $q_4$ , the packet error probability for the proposed ET-HARQ (shown in Fig. 2) can be obtained as in (13). Proof completes. ■

**Lemma 5:** The average complexity of ET-HARQ with CRC, when  $M = 2$ , is given by:

$$\begin{aligned} K_E \approx & K_{ET}^{(1)} + (1 - \epsilon_1)\alpha_1 K_s^{(1)} + \epsilon_1(1 - \beta_1)K_f^{(1)} \\ & + ((1 - \epsilon_1)(1 - \alpha_1) + \epsilon_1\beta_1 + \epsilon_1(1 - \beta_1)(1 - p_{ud})) K_{ET}^{(2)} \\ & + (1 - \epsilon_1)(1 - \alpha_1)\alpha_2 K_s^{(2)} \\ & + (\epsilon_1 - \epsilon_2)(\beta_1 + (1 - \beta_1)(1 - p_{ud})) \alpha_2 K_s^{(2)} \\ & + \epsilon_2(\beta_1 + (1 - \beta_1)(1 - p_{ud}))(1 - \beta_2)K_f^{(2)}, \end{aligned} \quad (14)$$

where  $K_{ET}^{(1)}$  and  $K_{ET}^{(2)}$  are respectively the complexity of the ET algorithm in the first and second transmission rounds.

*Proof:* We first focus on the first round of transmission. All packets received at the receiver will go through the ET process with complexity  $K_{ET}^{(1)}$ . The chance for a packet to

be decodable and declared decodable by the ET mechanism is  $(1 - \epsilon_1)\alpha_1$ . Such a packet undergoes decoding with complexity  $K_s^{(1)}$ . Moreover, the chance for a packet to be undecodable and falsely declared decodable by the ET mechanism is  $\epsilon_1(1 - \beta_1)$ . Such a packet undergoes decoding with complexity  $K_f^{(1)}$ .

Those packets that are declared undecodable by the ET mechanism in the first round of transmission or have undergone decoding and CRC check failed, will go through the second round of transmission and accordingly the second ET process with complexity  $K_{ET}^{(2)}$ . In fact, a packet will go through the second round of transmission, if it is decodable but was declared undecodable by the ET mechanism (this occurs with probability  $(1 - \epsilon_1)(1 - \alpha_1)$ ), if it is undecodable and was declared undecodable by the ET mechanism (which occurs with probability  $\epsilon_1\beta_1$ ), or if it is undecodable but was declared decodable by the ET mechanism and after the decoding, the CRC checks correctly detected the unsuccessful decoding (which occurs with probability  $\epsilon_1(1 - \beta_1)(1 - p_{ud})$ ). Therefore, the probability that a packet will be scheduled for the second round of transmission is  $(1 - \epsilon_1)(1 - \alpha_1) + \epsilon_1\beta_1 + \epsilon_1(1 - \beta_1)(1 - p_{ud})$ , and accordingly, the average complexity contributed by the ET mechanism in the second round of transmission is given by the 2nd line of (14).

If a packet is decodable in the first round but is declared undecodable by the ET mechanism (this occurs with probability  $(1 - \epsilon_1)(1 - \alpha_1)$ ), it is declared decodable in the second round with probability  $\alpha_2$  and is decoded successfully with complexity  $K_s^{(2)}$ . In such a scenario, the average complexity is  $(1 - \epsilon_1)(1 - \alpha_1)\alpha_2 K_s^{(2)}$ , which is equivalent to the 3rd line of (14). Moreover, if a packet is undecodable in the first round and is scheduled for the second round of transmission (which occurs with probability  $\epsilon_1(\beta_1 + (1 - \beta_1)(1 - p_{ud}))$ ), it can be decoded in the second round of transmission with probability  $1 - \epsilon'_2$ , only if it is declared decodable by the ET mechanism, which occurs with probability  $\alpha_2$ . In such a scenario, the average decoding complexity will be  $\epsilon_1(\beta_1 + (1 - \beta_1)(1 - p_{ud}))\alpha_2(1 - \epsilon'_2)K_s^{(2)}$ , which is equivalent to the line 4 of (14), by assuming  $\epsilon'_2\epsilon_1 \approx \epsilon_2$  according to (4).

Finally, if a packet is undecodable in the first round and is scheduled for the second round of transmission, but remained undecodable in the second round (which occurs with probability  $\epsilon_1(\beta_1 + (1 - \beta_1)(1 - p_{ud}))\epsilon'_2$ ), it is falsely declared decodable with probability  $1 - \beta_2$ , and undergone decoding with complexity  $K_f^{(2)}$ . In this case, the average complexity is  $\epsilon_1(\beta_1 + (1 - \beta_1)(1 - p_{ud}))\epsilon'_2(1 - \beta_2)K_f^{(2)}$ , which is equivalent to the last line of (14) using the assumption (4). ■

**Remark 2:** The throughput of ET-HARQ with  $k_C$ -bit CRC, when  $M = 2$ , is given by

$$\eta_E \approx k_I \frac{(1 - \zeta_E)}{K_E T_b}, \quad (15)$$

where  $k_I = k - k_C$  and  $T_b$  is the time required for a binary operation on the decoder hardware platform.

### B. ET-HARQ With CRC for $M$ Transmission Rounds

We now extend the analysis to ET-HARQ with  $M$  transmission rounds. Similar to the analysis in Section III-B, for round

$m$ , two key probabilities determine the system behavior: the probability of transmission termination  $p_m^t$  and the probability of requesting retransmission  $p_m^r$ .

The transmission termination probability in round  $m$  is

$$p_m^t = p_{m-1}^t [\alpha_m(1 - \epsilon_m) + (1 - \beta_m)\epsilon_m p_{ud}], \quad (16)$$

where the first term represents correct prediction of decodable packets ( $\alpha_m$ ) that are actually decodable ( $1 - \epsilon_m$ ), and the second term accounts for incorrectly predicted decodable packets ( $1 - \beta_m$ ) that are undecodable ( $\epsilon_m$ ) but pass CRC check ( $p_{ud}$ ).

The probability of requesting the  $(m + 1)$ -th transmission is

$$p_m^r = p_{m-1}^r [(1 - \alpha_m)(1 - \epsilon_m) + \beta_m\epsilon_m + (1 - \beta)\epsilon_m(1 - p_{ud})], \quad (17)$$

comprising three scenarios: missed detection of decodable packets, correct prediction of undecodable packets, and incorrect prediction of undecodable packets that fail CRC check. For  $m = 1$ , these probabilities initialize as  $p_1^t = \alpha_1(1 - \epsilon_1) + (1 - \beta_1)\epsilon_1 p_{ud}$  and  $p_1^r = (1 - \alpha_1)(1 - \epsilon_1) + \beta_1\epsilon_1 + (1 - \beta)\epsilon_1(1 - p_{ud})$ .

Based on the above recursive relationship, the PER of ET-HARQ for general  $M$  rounds can be obtained as

$$\zeta_E(M) = (1 - \beta_1)\epsilon_1 p_{ud} + \sum_{m=2}^{M-1} p_{m-1}^r (1 - \beta_m)\epsilon_m p_{ud} + p_{M-1}^r [(1 - \alpha_m)(1 - \epsilon_m) + \epsilon_m], \quad (18)$$

where the terms represent undetected errors in the first transmission, intermediate rounds, and final transmission respectively.

The average complexity considers both successful and failed decoding attempts weighted by their occurrence probabilities as

$$\begin{aligned} K_E(M) &= K_s^{(1)}(1 - \alpha_1)(1 - \epsilon_1) + K_f^{(1)}(1 - \beta_1)\epsilon_1 \\ &+ \sum_{m=2}^M p_{m-1}^r (K_s^{(m)}(1 - \alpha_m)(1 - \epsilon_m) + K_f^{(m)}(1 - \beta_m)\epsilon_m). \end{aligned} \quad (19)$$

The first two terms account for initial transmission complexity, while the summation captures subsequent rounds where decoding occurs only for missed detections ( $1 - \alpha_m$ ) of decodable packets and false predictions ( $1 - \beta_m$ ) of undecodable packets.

Finally, the throughput can be obtained as

$$\eta_E(M) \approx k_I \frac{(1 - \zeta_E(M))}{K_E(M)T_b}. \quad (20)$$

### C. Selective ET-HARQ

In the proposed ET-HARQ, as presented in Fig. 2, the received packets pass through the ET mechanism in each transmission round, i.e.,  $m = 1, 2, \dots, M$ . Therefore, if the packet is found not decodable in all rounds by the ET, its decoding process is eliminated altogether. This can result in the loss of decodable packets if the receiver terminates its decoding



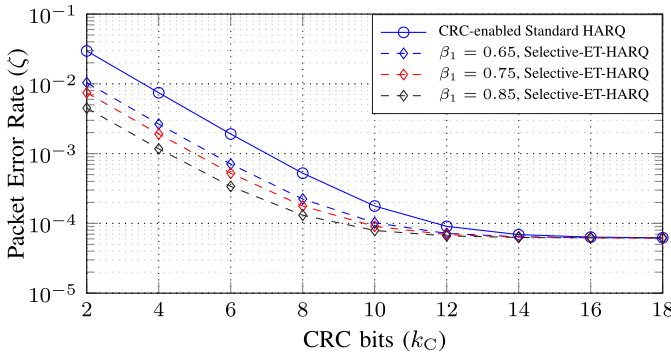


Fig. 4. Impact of CRC length on the PER at SNR=3dB.

due to false prediction in all rounds. We can modify the ET-HARQ to avoid the loss of these decodable packets by always performing full decoding in the last round of transmission. We refer to this scheme as selective ET-HARQ. The selective ET-HARQ targets to reduce the decoding complexity, without compromising the reliability performance.

**Lemma 6:** The PER, average complexity, and throughput of selective ET-HARQ can be obtained from (18), (19), (20) respectively, by setting  $\alpha_M = 1$ ,  $\beta_M = 0$ , and  $K_{ET}^{(M)} = 0$ .

*Proof:* Setting  $\alpha_M = 1$  implies that all the received packets in the last transmission round will always go for full decoding. Thus, setting  $\beta_M = 0$ , which implies that no packet is dropped from full decoding in the second round, is also required. Moreover, since we avoid ET in the last round, therefore, for average complexity of selective ET-HARQ  $K_{ET}^{(M)} = 0$ . ■

## V. NUMERICAL ANALYSIS

### A. Impact of ET With CRC

The CRC check performs two critical functions in standard HARQ: error detection and transmission termination. However, this approach requires full ML decoding before making retransmission decisions, introducing significant computational overhead for each transmission attempt. Our proposed ET mechanism operates prior to ML decoding, complementing rather than replacing the CRC check.

The effectiveness of standard HARQ strongly depends on CRC length. With shorter CRC lengths, the probability of undetected errors ( $p_{ud}$ ) increases, causing more failed packets to be mistakenly accepted during initial transmissions. These undetected errors result in missed retransmission opportunities and degraded system reliability. The ET mechanism helps mitigate this limitation by providing an additional layer of verification before committing to full decoding. By identifying likely undecodable packets early, ET reduces unnecessary decoding attempts while preserving the error detection capabilities of CRC.

Fig. 4 depicts the PER variation of the CRC-enabled standard HARQ and the selective ET-HARQ under the influence of CRC length. Notably, the PER performance of the selective ET-HARQ is better than that of the only CRC-enabled standard HARQ for short to medium CRC lengths. This is because the selective ET-HARQ does not solely rely on the CRC check but is also influenced by the accuracy of ET prediction for

calling retransmissions. According to Lemma 6, the PER of selective ET-HARQ is given as

$$\begin{aligned} \zeta_E &\approx \epsilon_1(1 - \beta_1)p_{ud} + \epsilon_2((1 - \beta_1)(1 - p_{ud}) + \beta_1) \\ &= \epsilon_1(1 - \beta_1)p_{ud} + \epsilon_2(1 - \beta_1) - \epsilon_2(1 - \beta_1)p_{ud} + \epsilon_2\beta_1 \\ &= p_{ud}(1 - \beta_1)(\epsilon_1 - \epsilon_2) + \epsilon_2. \end{aligned} \quad (21)$$

If  $\beta_1 = 0$ , the PER (21) of selective ET-HARQ reduced to the PER of the CRC-enabled standard HARQ given in (5). Therefore, any selective ET-HARQ with  $\beta_1 > 0$  can mitigate the reliability loss introduced by undetected errors of short CRC. Consequently, when the ET prediction accuracy  $\beta_1$  is higher, the PER performance of the selective ET-HARQ improves, as shown in Fig. 4. Finally, when an adequate number of CRC bits are provided leading  $p_{ud} \rightarrow 0$ , both the selective ET-HARQ and the standard HARQ exhibit the same PER reliability. This feature of selective ET-HARQ will be further verified by simulations in Section VI.

### B. PER, Decoding Complexity and Throughput

In this section, we compare the proposed ET-HARQ and the standard HARQ schemes in terms of reliability, complexity, and throughput. For this, we fixed CRC at  $k_C = 8$  bits for all the numerical simulations and focused on the impact of the accuracy of the ET mechanism. Note that the analytical evaluation of the HARQ schemes in the previous sections, i.e., III and IV, is general and can accommodate any encoding and decoding procedure. However, in this section, we specifically consider BCH codes and the OSD decoder for numerical comparison purposes.

When OSD exhausts all test error patterns (TEP), it leads to the worst decoding complexity, denoted as  $K_f^m(s)$ . This situation occurs when the packet fails. However, we employ the Probability-Based OSD (PB-OSD) algorithm which incorporates an internal early termination mechanism. PB-OSD can dynamically terminate the search once a reliable codeword is found [37], resulting in significantly reduced average complexity  $K_s^m(s)$ . We note that  $K_s^m(s)$  depends on the SNR and order  $s$ , and  $K_s^m(s) \ll K_f^m(s)$ . Appendix briefly introduces OSD and gives further details on the decoding complexities. Although PB-OSD has an internal early termination mechanism, our proposed ET is further applied before the decoding process.

The analysis until now is general and does not specify the exact ET method. In Section VI, we design a practical ET mechanism having much lower complexity than the decoding process, i.e.,  $K_{ET}^{(m)} \ll K_s^{(m)}$ , and sufficient prediction accuracy. Section VI-D will show that the proposed NN-based ET further reduces the complexity of PB-OSD with introducing minimal extra overhead.

In numerical analysis of this section, we set the exact complexity of ET, i.e.,  $K_{ET}^{(1)} = 15632$ , and  $K_{ET}^{(2)} = 60752$  based on the proposed NN-based ET mechanism presented in Section VI. First, we show how an ET mechanism with even a modest prediction accuracy can yield notable complexity improvements with minimal reliability degradation.

In Fig. 5, we examine the performance of ET-HARQ in terms of PER, throughput, and decoding complexity, with low



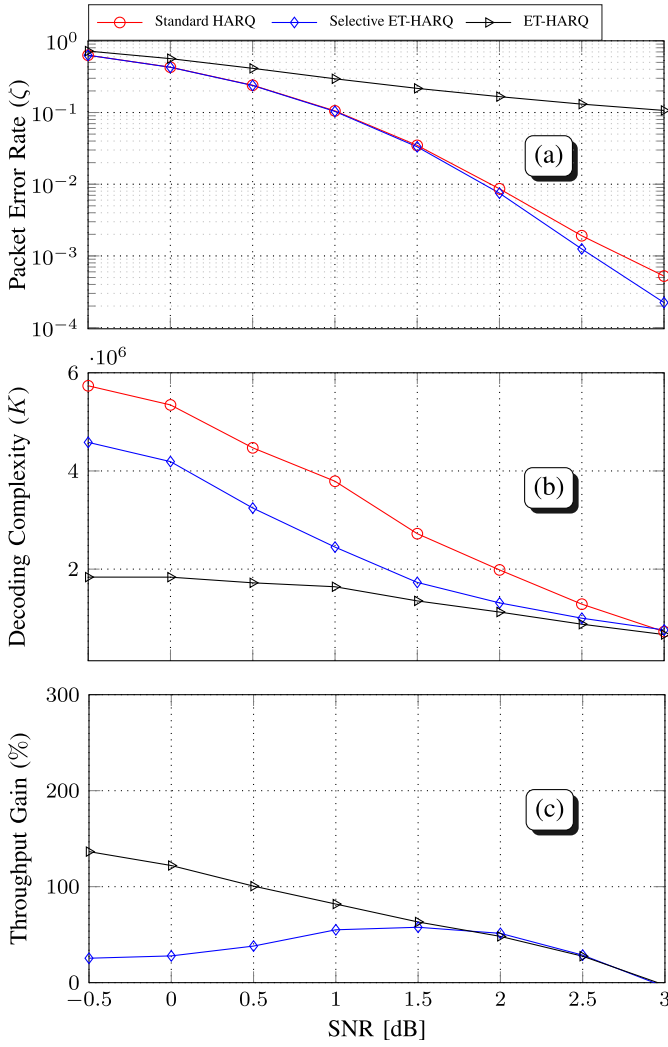


Fig. 5. Performance comparison of standard HARQ and ET-HARQ for  $\alpha_1 = \beta_1 = 0.65$ ,  $\alpha_2 = \beta_2 = 0.75$ ,  $k = 64$ ,  $n_1 = 96$ ,  $n_2 = 128$ , and  $k_C = 8$ .

and moderate ET prediction accuracy. Setting  $\alpha_1 = \beta_1 = 0.65$ ,  $\alpha_2 = \beta_2 = 0.75$  reflect low prediction accuracy, whereas setting  $\alpha_1 = \beta_1 = 0.75$ ,  $\alpha_2 = \beta_2 = 0.85$  reflect moderate prediction accuracy. Note that in this Section, we keep the ET prediction accuracy fixed across all SNRs for general analysis and simulation purposes. However the exact trend of ET prediction accuracy depends on specific mechanisms and in reality, it can vary with SNR as well.

Fig. 5(a) shows the PER comparison of various HARQ schemes with low prediction accuracy settings. Comparing ET-HARQ and selective ET-HARQ, when ET prediction is applied in both rounds, the PER performance is worse than that of standard HARQ. This is due to a higher chance of successful packets being lost due to low ET prediction accuracy. However, the PER of selective ET-HARQ is not impacted by the loss of decodable packets in the first round because they are eventually recovered in the last round due to OSD decoding. We also observe that the PER performance of selective ET-HARQ is gradually improving compared to standard HARQ from SNR of 1.5 dB. This is due to an imperfect CRC check, which impacts the standard HARQ more than selective ET-HARQ, as explained in Sec. V-A.

Fig. 5(b) illustrates the decoding complexity comparison among different HARQ techniques in the low ET prediction accuracy. Notably, a significant reduction in complexity is observed with ET-HARQ compared to standard HARQ. This reduction is particularly pronounced in low to medium SNR regimes, where retransmissions are more frequent. Part of the complexity reduction with ET-HARQ is due to avoiding the decoding of many packets, including some decodable ones, which is evident in the loss of PER performance, as explained earlier. However, the selective ET-HARQ also exhibits substantial complexity reduction compared to standard HARQ. Importantly, the complexity performance gain of selective ET-HARQ comes without any loss in PER performance. On the contrary, with selective ET-HARQ, we see some improvement in PER in the high SNR ranges. This indicates that the proposed ET-based retransmission offers fundamental performance compared to the retransmission mechanism adopted in standard HARQ. Since the complexity gain is mainly due to efficiently conducting retransmissions, a reduced probability of retransmission with increasing SNR gradually decreases the complexity gain in all HARQ schemes.

Next, we present the comparison of throughput. Since the bit processing time  $T_b$  depends on hardware capabilities, we show the relative throughput gain of ET-HARQ with respect to standard HARQ as

$$\Delta\eta_E = \frac{(\eta_E - \eta_S)}{\eta_S}, \quad (22)$$

where  $\eta_S$  and  $\eta_E$  are defined in (7) and (15), respectively.

As depicted in Fig. 5(c), both ET-HARQ and selective ET-HARQ demonstrate significant throughput gains compared to standard HARQ. ET-HARQ particularly shows remarkable throughput gains, especially in the low and medium SNR regimes. Specifically, ET-HARQ maintains above 100% throughput gain between SNR -0.5 dB and 0.5 dB. However, this improvement is primarily due to a substantial reduction in decoding complexity rather than an improvement in PER, as explained earlier. As SNR increases, the throughput gain of ET-HARQ declines, reaching approximately 55% at 2 dB SNR. In contrast, selective ET-HARQ shows a gradual increase in throughput gain from around 25% at -0.5 dB SNR. Interestingly, at 2 dB SNR, the throughput gains of ET-HARQ and selective ET-HARQ converge closely to approximately 55%. Selective-ET-HARQ's advantage lies in its ability to expedite retransmissions and minimize decoding complexity without dropping decodable packets, leading to notable throughput enhancements in medium SNR ranges. Finally, at high SNRs, the throughput gains diminish due to fewer retransmissions.

In Fig. 6, we explore the impact of increasing the ET accuracy on PER, decoding complexity, and throughput performance of various HARQ schemes in a moderate prediction accuracy setting. The overall performance trends remain intact, while we see further performance enhancement with ET-HARQ and selective ET-HARQ in comparison with standard HARQ. For example, Fig. 6(a) indicates marginal improvement in the PER performance of ET-HARQ. This is because, with higher ET accuracy, undecodable failing packets in the

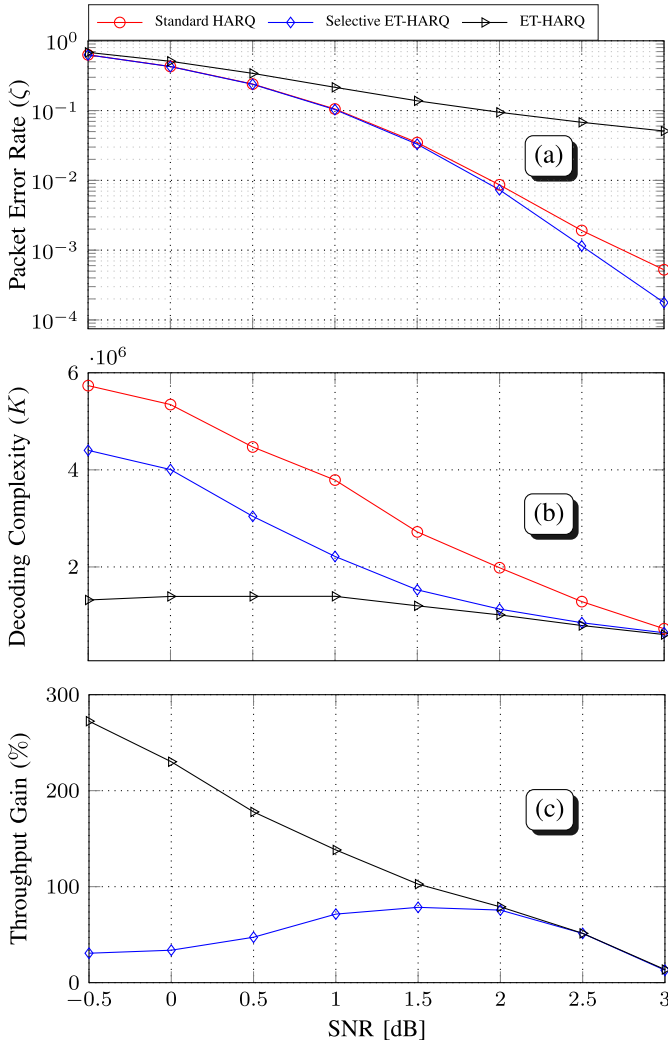


Fig. 6. Performance comparison of standard HARQ and ET-HARQ for  $\alpha_1 = \beta_1 = 0.75$ ,  $\alpha_2 = \beta_2 = 0.85$ ,  $k = 64$ ,  $n_1 = 96$ ,  $n_2 = 128$ , and  $k_C = 8$ .

first round are now better identified for retransmission as well as undetected decodable packets are better identified for decoding. Note that the PER of selective ET-HARQ barely improves it is already close to the maximum due to allowed full decoding in the last round.

Fig. 6(b) shows decoding complexity performance with increased ET accuracy. Comparing Fig. 5(b) and Fig. 6(b), it can be seen that both ET-HARQ and selective ET-HARQ exhibit further performance improvement in comparison with standard HARQ due to better ET accuracy. Specifically, the ET-HARQ exhibits notable decoding complexity reduction in Low and medium SNR regimes, whereas selective ET-HARQ shows prominent complexity reduction in medium SNR regimes in comparison to the decoding complexity of baseline standard HARQ. This is because with better ET accuracy more undecodable packets can be correctly called for retransmissions without performing full decoding, leading to further average decoding complexity reduction.

Fig. 6(c) indicates the throughput gain of ET-HARQ schemes relative to standard HARQ with increased ET accuracy. Comparing Fig. 5(c) and Fig. 6(c), we see the throughput gain of the ET-HARQ almost doubled in low SNR regimes due

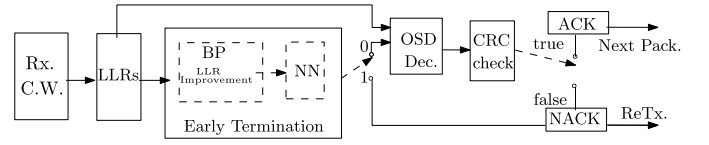


Fig. 7. Belief-Propagation Neural-Network based Early Termination.

to slight increase in ET accuracy. For selective ET-HARQ, although the throughput gain increases in all SNR regimes, we see very high throughput gain between SNR 1 dB to 2 dB. For example, the throughput gain of selective ET-HARQ is approximately 79% more than standard HARQ at 2 dB SNR. Finally, it is concluded that with ET-HARQ, even with modest early termination accuracy, substantial reductions in complexity can be achieved resulting in huge through gains without compromising PER significantly.

## VI. NEURAL NETWORK-BASED EARLY TERMINATION

In the previous section, we derive the analytical expressions involving general ET prediction parameters, i.e.,  $\alpha_m$ ,  $\beta_m$ , etc. This highlights how HARQ schemes can benefit in terms of complexity through the ET mechanism. Here, we present a neural network (NN)-based ET mechanism for predicting decodable and undecodable packets. We demonstrate that even a small NN can achieve satisfactory ET accuracy and significantly enhance ET-HARQ performance.

Fig. 7 shows the block diagram of the proposed scheme, where a pre-trained artificial NN is used to classify the incoming codeword as decodable or undecodable, indicated by 0 and 1, respectively. The statistics of channel log-likelihood ratios (LLRs) are taken as input to the classifier. Moreover, we use a few iterations of belief propagation (BP) decoding to improve the raw LLRs as the additional classifier input.

Note that the BP is not used for decoding but only to enhance the NN performance, which should be distinguished from concatenated decoder designs such as in [36]. The ET mechanism involves offline training with labeled data. For this purpose, various codewords are generated randomly at specific SNRs. After the transmission over the AWGN channel, the input features are derived from the statistics of channel LLRs in conjunction with BP results. These features and the corresponding decoding outcomes obtained by an actual decoder are recorded as labels after the decoding process outcome. A NN is trained using these samples to predict the decodability of received noisy codewords, enabling its deployment for real-time decision-making scenarios.

### A. Preparation of Input Features

Input features are generated using a BP decoder, which is a widely acknowledged iterative decoding method for low-density parity-check (LDPC) codes. BP can be applied to any linear block codes represented by a Tanner graph. More details on the BP algorithm and Tanner graph can be found in [38] and [39]. While BP achieves near-optimal error rates with LDPC codes, it is less effective with high-density codes like BCH, primarily due to the presence of small cycles in their Tanner graphs [40]. Moreover, the computational complexity of BP scales linearly with the length of the codeword [38].

In the  $m$ -th transmission round, let the channel LLRs of the observed symbols  $\mathbf{y}_m$  be denoted by  $\ell_m = [\ell_1, \ell_2, \dots, \ell_{n_m}]$ , where the  $i$ -th LLR is given by  $\ell_i = 2\gamma y_i$  for Binary Phase Shift Keying (BPSK) symbols. During the decoding, BP iteratively updates the LLRs between variable nodes and checks nodes along the edges of the Tanner graph. Let  $\mathcal{A}_i$  denote the set of check nodes connected to the  $i$ -th variable node and  $\mathcal{B}_j$  denote the set of variable nodes connected to the  $j$ -th check node. The message passing procedure in the  $t$ -th BP iteration is described as

$$M_{ji}^{(t)} = \ell_i + \sum_{j' \in \mathcal{A}_i \setminus j} E_{ij'}^{(t-1)}, \quad (23)$$

for all  $i \in \{1, \dots, n_m\}$  and  $j \in \mathcal{A}_i$ , and

$$E_{ij}^{(t)} = 2 \tanh^{-1} \left( \prod_{i' \in \mathcal{B}_j \setminus i} \tanh(M_{ji'}^{(t)}/2) \right), \quad (24)$$

for all  $j \in \{1, \dots, n_m - k\}$  and  $i \in \mathcal{B}_j$ . In (23) and (24),  $M_{ji}^{(t)}$  denotes the messages sent from the  $i$ th variable nodes to the  $j$ th check node, and  $E_{ij}^{(t)}$  denotes the messages returned from the  $j$ th check node to the  $i$ th variable node. Particularly,  $E_{ij}^{(0)} = 0$  for all  $i$  and  $j$ . After each iteration, BP computes the updated LLR  $\ell_m^{(t)} = [\ell_1^{(t)}, \dots, \ell_{n_m}^{(t)}]$  according to

$$\ell_i^{(t)} = \ell_i + \sum_{j \in \mathcal{A}_i} E_{ij}^{(t)}, \quad (25)$$

which is regarded as the posterior LLR of  $y_i$ . Henceforth, the prior LLR  $\ell_m$  is denoted by  $\ell_m^{(0)}$  for consistency.

The trajectory of BP, i.e., changes of  $M_{ji}^{(t)}$ ,  $E_{ij}^{(t)}$ ,  $\ell_i^{(t)}$  or related variables across iterations, has been widely used in evaluating decoding performance, including the extrinsic-information-transform (EXIT) chart and density evolution analysis [41]. A fast-converging trajectory indicates reliable received symbols, while an earlier convergence point suggests better error performance. This motivates using BP trajectory as the input feature for our NN-based decodability classifier. Specifically, for a BP process totaling  $T$  iterations, the following quantities are included as NN inputs:

- First and second-order moments of  $\ell_1^{(t)}, \dots, \ell_{n_m}^{(t)}$ .
- Information density computed from  $\ell_1^{(t)}, \dots, \ell_{n_m}^{(t)}$ .
- First-order moment of  $|\ell_1^{(t)}|, \dots, |\ell_{n_m}^{(t)}|$ .

The information density for binary AWGN channel is computed as [42]

$$\iota = \sum_{i=1}^{n_m} 1 - \log(1 + \exp(-x_i \ell_i^{(t)} / \gamma)),$$

where  $x_i$  is the hard decision of  $\ell_i^{(t)}$ . These quantities are calculated for iterations  $t = 0, \dots, T$ . Thus, the total number of input features is  $4(T+1)$ .

### B. Neural Network-Based Early Termination Classifier

The input configuration for the NN-based ET mechanism is described in Section VI-A. The output of the ET is a binary decision indicating whether the observed  $\mathbf{y}_m$  is decodable or undecodable. The architecture consists of four hidden layers,

TABLE I  
PARAMETERS OF THE NN-BASED ET SYSTEM

|               |         |               |               |
|---------------|---------|---------------|---------------|
| Optimizer     | Adam    | Learning rate | $10^{-4}$     |
| Hidden layers | 4       | Hidden size   | 64            |
| Activation    | ReLU    | Batch size    | 512           |
| BP iterations | $T = 5$ | NN complexity | $O(NH + H^2)$ |

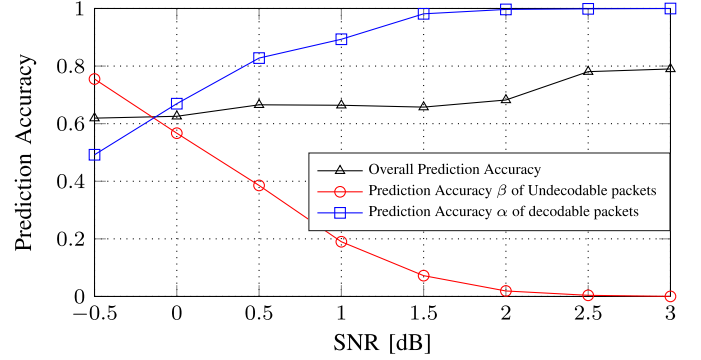


Fig. 8. Prediction accuracy of the NN-based ET for the (128, 64) eBCH code.

each equipped with 64 neurons. The activation function used in all layers is the Rectified Linear Unit (ReLU). Drawing inspiration from the work in [43], our model incorporates residual shortcut connections across its layers. This innovative approach effectively counters the challenges of vanishing gradients and performance decline typically associated with increasing network depth.

The model is trained individually for each SNR level, ensuring a balanced training set with an equal number of decodable and undecodable packets. The training set size for each SNR level is set to  $10^5$ . To normalize each SNR's input features, we subtract the mean and divide it by the feature's variance across the dataset. Subsequently, the training dataset is organized into mini-batches of 512 samples each. We utilize the ADAM optimization algorithm, initializing it with a learning rate of  $10^{-4}$ .

Moreover, we incorporate an early stopping mechanism to improve training efficiency. This mechanism halts the training process if no performance improvement is detected on the validation set, as measured against the moving average of the previous five epochs. This strategy ensures the model's convergence while mitigating overfitting. The parameters of the NN-based ET mechanism are summarized in Table I.

The training and validation accuracy for the (128, 64) extended Bose-Chaudhuri-Hocquenghem (eBCH) code is depicted in Fig. 8. The prediction accuracy for actual decodable and undecodable is different and also varies with SNR. At low SNR, i.e.,  $-0.5$  dB, NN-based ET exhibits  $\approx 50\%$  accuracy in identifying actual decodable packets. Whereas it correctly identifies an actual undecodable packet with accuracy  $\approx 75\%$ . The correct prediction accuracy for actual decodable packets improves significantly with SNR and reaches close to 100% in high SNR regimes. This is because high SNR regimes better outline the features indicating decodable packets and vice versa. On the other hand, the prediction accuracy for undecodable packets is high in a low SNR regime and gradually reduces with an increase in SNR. At very high



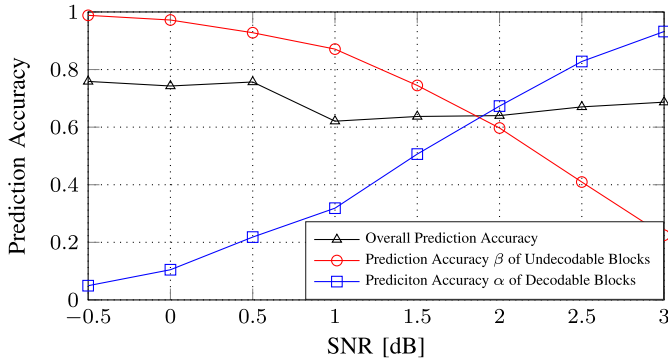


Fig. 9. Prediction accuracy of the NN-based ET Algorithm for a (96, 64) code punctured from the (128, 64) eBCH code.

SNRs, e.g., 2.5 dB and 3 dB, no received packets are inferred as undecodable, resulting in no recorded outcome for them. The overall prediction accuracy significantly improves with increasing SNR.

By increasing the code rate, the prediction accuracy is expected to decrease. To demonstrate this, the training and validation accuracy for a (96, 64) code is illustrated in Fig. 9. The code is obtained by optimally puncturing the parity bits of (128, 64) eBCH codewords, where the code minimum Hamming distance is maximized. The prediction accuracy trends are similar for actual decodable and undecodable packets, and the overall accuracy for the (96, 64) code is around 70%

### C. Computational Complexity of the NN-Based ET

According to [44], computing (23) or (25) requires an average of  $n_m(n_m - k)$  additions,  $8n_m(n_m - k - 1)$  multiplications, and  $n_m(n_m - k)$  divisions. Computing (24) requires an average of  $2n_m(n_m - k)$  additions and  $3n_m(n_m - k - 1)$  multiplications. Therefore, the complexity of BP is approximately represented as  $O(n_m(n_m - k))$ , which increases linearly with the codeword length  $n_m$ .

In the NN-based ET prediction, the inputs layer takes  $N$  dimensional input and maps it to a hidden layer with dimension  $H$ . Here, the primary computational burden is a matrix multiplication involving an input vector of length  $N$  and a weight matrix with dimensions  $N \times H$ . Consequently, the computational complexity is  $O(NH)$ . Similarly, the computational complexity between hidden layers and from the last hidden layer to the output layer is  $O(H^2)$  and  $O(H)$ , respectively. Thus, the total computational complexity for a forward pass in our architecture is given by  $O(NH + H^2)$ . Overall, the complexity of the designed ET mechanism is  $K_{ET}^m \approx TO(n_m(n_m - k)) + O(NH + H^2)$ .

Note that compared to the complexity of a typical near ML decoding algorithm, the complexity of the proposed NN-based ET mechanism is negligible. For example, in Appendix, it's detailed that the complexity of OSD is significantly higher due to a large number of test error patterns, as shown in equation (26). However, for very short codes, the ET overhead becomes more prominent relative to their efficient ML decoding operations, particularly at higher SNRs. This results in diminishing complexity and throughput gains at higher SNRs, as will be observed in simulations in Section VI-D.

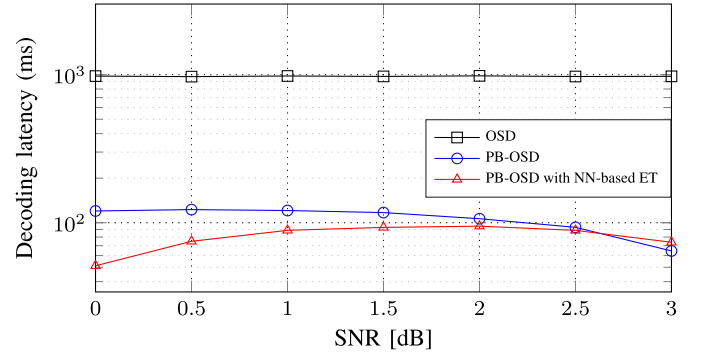


Fig. 10. Prediction accuracy of the NN-based ET Algorithm for a (96, 64) code punctured from the (128, 64) eBCH code.

The proposed ET mechanism is designed with hardware implementation feasibility. The BP computation can be efficiently implemented using fixed-point arithmetic with minimal precision loss [45]. Our NN architecture is intentionally kept simple, with only four hidden layers of 64 neurons each, making it suitable for FPGA implementation. Similar lightweight neural networks have been successfully implemented on FPGAs [46], achieving processing latencies in the order of microseconds. While detailed hardware implementation is beyond the scope of this paper, these considerations demonstrate the practical feasibility of our approach.

*Remark 3:* We acknowledge that training separate NN-based ET for different code lengths and SNRs introduces extra storage and training overhead, particularly in HARQ. However, our ET neural network is intentionally minimal with only 15K parameters (4 hidden layers with 64 neurons each as outlined in Table I with  $N \times H + 3H^2 + H$  parameters), resulting in a storage requirement of merely 20KB per model and training time of less than 10 minutes per code length on a standard GPU. This makes the approach practical for URLLC systems that typically operate with a limited set of standardized code lengths. For applications requiring broader scalability across arbitrary code lengths, advanced techniques such as Graph Neural Networks (GNN) and meta-learning offer promising solutions. GNNs can leverage the inherent graph structure of linear block codes to process different code lengths with a single set of parameters [47], while meta-learning enables rapid adaptation to new code lengths with minimal fine-tuning samples [48]. These extensions are left for future work particularly for ET design.

### D. Simulation Results

1) *Complexity of the Proposed ET in Single Round Decoding:* We analyze the decoding latency for a single transmission round using three schemes: (i) standard OSD without any ET, (ii) PB-OSD and (iii) PB-OSD with our external BP-NN-based ET. Throughout this paper, all simulations are conducted using Python 3.8 on an i7-10700 CPU with NVIDIA 3090 GPU. The latency measurements are obtained by processing 10,000 blocks and computing the average time per block to ensure statistical accuracy. Here we focus on decoding (96, 64) eBCH codes using OSD order 4.

As shown in Fig. 10, the three schemes exhibit distinct latency behaviors. Standard OSD maintains a constant latency

TABLE II  
LATENCY BREAKDOWN (MS) FOR DIFFERENT DECODING SCHEMES

| SNR (dB) | OSD   | PB-OSD | PB-OSD with ET |          |       |
|----------|-------|--------|----------------|----------|-------|
|          |       |        | ET             | Decoding | Total |
| 0.0      | 980.2 | 120.3  |                | 45.4     | 50.9  |
| 0.5      | 980.2 | 122.1  |                | 68.3     | 73.8  |
| 1.0      | 980.2 | 119.7  |                | 82.6     | 88.1  |
| 1.5      | 980.2 | 117.2  | 5.5            | 86.3     | 91.8  |
| 2.0      | 980.2 | 106.4  |                | 88.7     | 94.2  |
| 2.5      | 980.2 | 92.8   |                | 82.3     | 87.8  |
| 3.0      | 980.2 | 64.1   |                | 67.4     | 72.9  |

of approximately 980 ms across all SNRs, as it exhaustively searches through all 679,121 TEPs (see Appendix) regardless of channel conditions. PB-OSD dramatically reduces this latency through its probabilistic early stopping, achieving latencies between 64–122 ms that decrease with increasing SNR. Our proposed BP-NN-based ET further reduces the decoding latency at low SNRs, achieving 51 ms at 0 dB SNR.

Notably, there exists a crossover point at approximately 2.5 dB where PB-OSD alone becomes more efficient than PB-OSD with NN-based ET. As highlighted in Table II, this occurs because our ET mechanism introduces a fixed overhead of 5.523 ms per block. At low SNRs, the ET effectively identifies and skips the majority of undecodable blocks to decrease the overall decoding latency. However, at high SNRs, most blocks become readily decodable, reducing the benefit of NN-based ET before decoding. The fixed 5.5 ms overhead thus becomes dominant relative to the marginal gains.

2) *BCH Codes*: We conduct the simulation to validate the performance of ET-HARQ with an NN-based ET mechanism. The receiver structure is shown in Fig. 7. Considering a two-round transmission, the initial transmission uses (96, 64) code, and the potential re-transmission uses (128, 64) eBCH code. The (96, 64) code results from puncturing the parity bits of the (128, 64) code; thus, they share the same information bits.

The probability-based OSD (PB-OSD) algorithm with OSD order 4 is applied for the OSD decoding module of Fig. 7 [37]. The PB-OSD is briefly introduced in the Appendix, and its complexity is measured by (29). The complexity of NN-based ET mechanism is measured by  $TO(n_m(n_m - k)) + O(NH + H^2)$  according to Section VI-C. Therefore, the overall complexity is measured by combining the complexities of PB-OSD and the NN-based ET for all potential decoding rounds. We apply CRC bits of length  $k_C = 8$  in the simulation, i.e.,  $p_{ud} = 1/256$ .

Fig. 11(a) shows that the selective ET-HARQ outperforms CRC-enabled standard HARQ in PER performance, as predicted by the analysis. As shown in Fig. 8, the ET-HARQ experiences significant PER degradation due to the low prediction accuracy of decodable packets, mainly due to mistakenly avoiding decoding of decodable packets in both rounds.

The complexity of standard HARQ, ET-HARQ, and the selective ET-HARQ is compared in Fig. 11(b). As illustrated, selective ET-HARQ exhibits lower complexity than the standard HARQ, whereas ET-HARQ has the lowest complexity since it can skip the decoding process for non-decodable received symbols in the last packet transmission round, as predicted by the analysis. Using the simulation-based PER and the

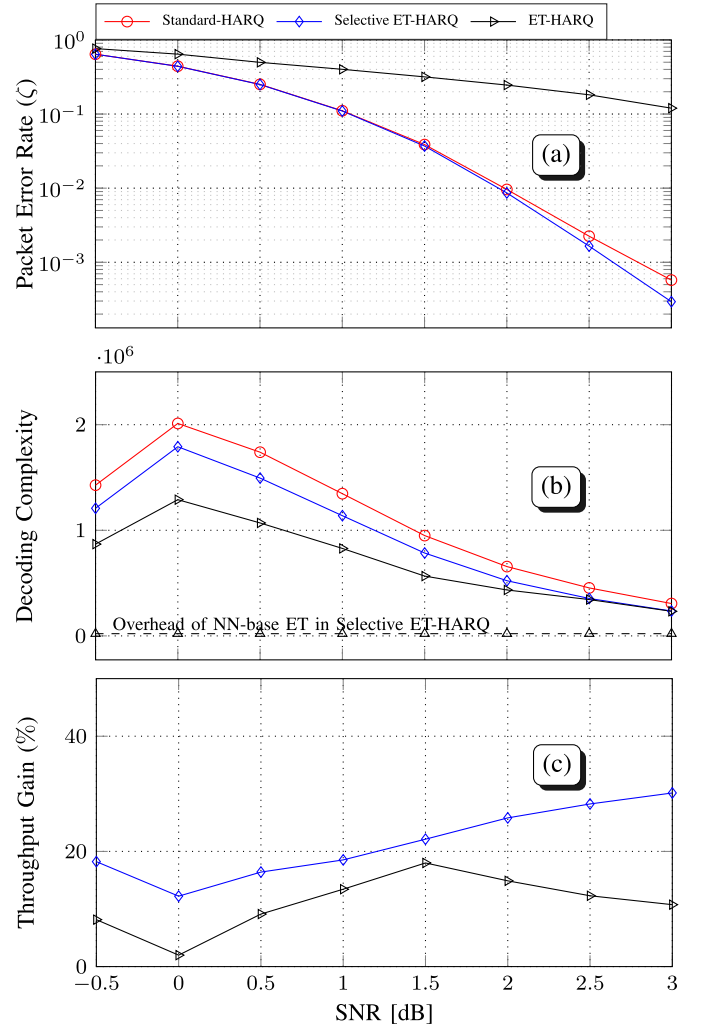


Fig. 11. Performance comparison of standard HARQ and ET-HARQ, when  $k = 64$ ,  $n_1 = 96$ ,  $n_2 = 128$ ,  $k_C = 8$ .  $\alpha_m$  and  $\beta_m$  values are vary at each SNR, due to BP NN-based ET according to Fig. 8 and 9.

decoding complexity, we obtain the relative throughput gain of ET-HARQ according to (22), as shown in Fig. 11(c). As shown in the figure, both ET-HARQ and selective ET-HARQ show high throughput gain in comparison with standard HARQ. We note that the complexity reduction and throughput gain are not as significant as the analytical results in Section V. We also illustrate the overhead of the NN-based ET in the selective ET-HARQ, which requires approximately  $2 \times 10^4$  operations per block, in contrast to the decoding complexity of over  $5 \times 10^5$  operations. Consequently, the overhead of NN-based ET is negligible relative to the overall decoding complexity.

Note that in the analysis, we keep the prediction accuracy fixed for all SNRs, which resulted in a high throughput gain in low SNRs. However, as shown in Figs. 8 and 9, the NN-based ET mechanism exhibits very low prediction accuracy for either undecodable or decodable blocks at very low or very high SNRs. Different from the analysis, here we also observe that selective ET-HARQ shows better throughput gain than ET-HARQ. This is again due to the difference between exact prediction accuracy variation with SNR in each round. Note that our focus here is to only show that the ET-HARQ will outperform the standard HARQ even with moderate prediction

TABLE III  
PREDICTION ACCURACIES FOR CRC-POLAR AT DIFFERENT SNRS

| SNR  | $n_1 = 32$ |           |      | $n_2 = 48$ |           |      | $n_3 = 64$ |           |      |
|------|------------|-----------|------|------------|-----------|------|------------|-----------|------|
|      | $\alpha_1$ | $\beta_1$ | Acc  | $\alpha_2$ | $\beta_2$ | Acc  | $\alpha_3$ | $\beta_3$ | Acc  |
| -3.0 | 30.6       | 84.9      | 67.0 | 30.6       | 84.9      | 62.2 | 76.2       | 30.9      | 69.9 |
| -2.0 | 48.5       | 76.1      | 65.3 | 48.5       | 76.1      | 65.9 | 89.8       | 14.1      | 70.2 |
| -1.0 | 65.2       | 63.3      | 64.2 | 65.2       | 63.3      | 67.0 | 97.4       | 4.8       | 80.7 |
| 0.0  | 83.3       | 47.2      | 65.8 | 83.3       | 47.2      | 70.8 | 99.5       | 0.8       | 77.0 |
| 1.0  | 94.9       | 29.3      | 67.1 | 94.9       | 29.3      | 78.9 | 99.8       | 0.1       | 80.9 |

accuracy. This however is visible even with a simple NN-based ET mechanism. One can apply more sophisticated NN models to achieve higher prediction accuracy, thereby reducing complexity further and increasing throughput.

3) *CRC-Polar Codes*: To further validate the effectiveness of ET-HARQ across different coding schemes, we evaluate its performance using rate-compatible 5G CRC-polar codes. For this evaluation, we consider a three-round transmission system with a short message length of  $k = 16$  bits and CRC length  $K_C = 6$  (i.e.,  $p_{ud} = 1/64$ ). The rate-compatible CRC-polar codes have blocklengths  $n_1 = 32$ ,  $n_2 = 48$ , and  $n_3 = 64$ , where the nested structure ensures that parity bits of higher-rate polar codes are contained within those of lower-rate polar codes. For each transmission round, we apply PB-OSD with order 3 to achieve near-ML decoding performance, with complexity comparable to successive cancellation list (SCL) algorithm for polar codes [12]. The neural network is trained separately for each blocklength to account for their distinct error patterns. The testing accuracy is summarized in Table III.

Figure 12 presents the performance comparison between standard HARQ and ET-HARQ schemes using these CRC-polar codes. As shown in Fig. 11(a), the PER performance exhibits similar trends to those observed with BCH codes, with selective ET-HARQ demonstrating superior reliability compared to standard HARQ. However, ET-HARQ shows notably higher PER, particularly in the low SNR regime. This performance degradation can be attributed to the increased challenge of accurately predicting decodable packets with very short polar codes during the final transmission round.

The complexity comparison is presented in Fig. 11(b). While both ET-HARQ variants still achieve complexity reduction at low SNRs compared to standard HARQ, these gains are less pronounced at high SNRs. This behavior stems from two key factors: first, the baseline decoding complexity of short polar codes ( $k = 16$ ) is much lower compared to the longer BCH codes ( $k = 64$ ) shown in Fig. 11; second, the computational overhead of the neural network-based ET mechanism becomes increasingly significant relative to the polar decoding complexity, particularly at higher SNRs where decoding operations are more efficient. The throughput gain analysis in Fig. 11(c) further illustrates this complexity-performance trade-off. Both ET-HARQ schemes demonstrate moderate throughput improvements in the low SNR regime, achieving approximately 20% gain. However, these benefits progressively diminish and ultimately become negative as SNR increases, primarily due to the NN overhead becoming dominant compared to the efficient polar decoding operations.

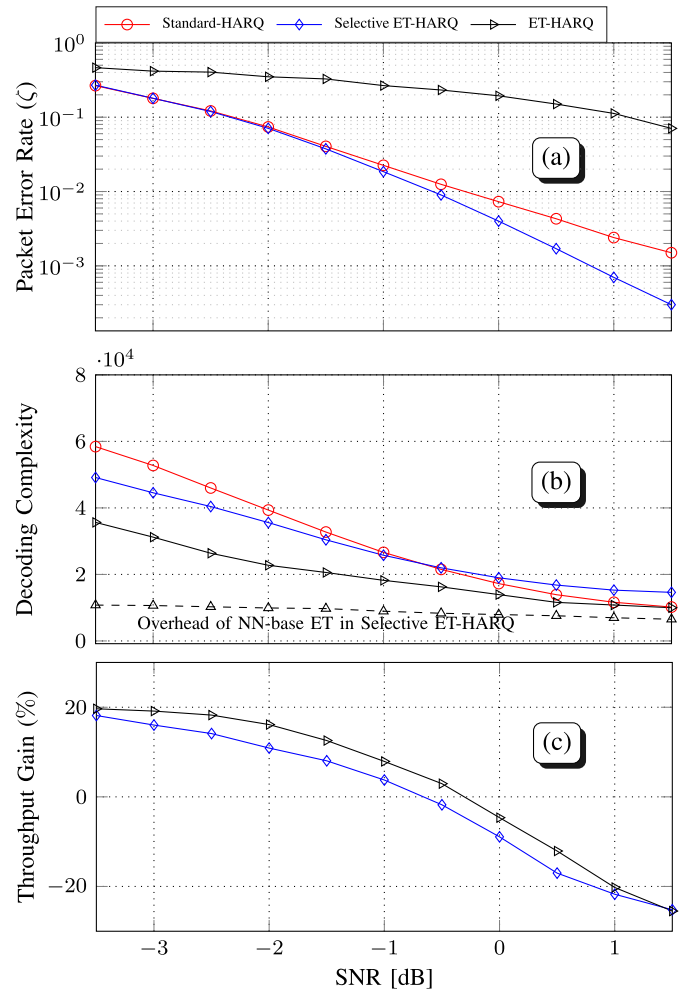


Fig. 12. Performance comparison of standard HARQ and ET-HARQ, using rate-compatible CRC-Polar codes ( $k_C = 6$ ) with  $k = 16$ ,  $n_1 = 32$ ,  $n_2 = 48$ , and  $n_3 = 64$ .

These results underscore a crucial design consideration: the complexity of the early termination mechanism must be carefully balanced against both the baseline decoding complexity and the code block length. The advantages of ET-HARQ may be less pronounced for codes with relatively efficient ML decoding, particularly at short block lengths.

## VII. CONCLUSION

We proposed and analyzed the ET-HARQ system in short-packet communication in the context of URLLC scenarios. ET-HARQ effectively identifies suitable packets for retransmission before decoding. We characterized the impact of ET-HARQ on PER, decoding complexity, and throughput with a practical CRC-enabled HARQ system to show its efficacy. Our results demonstrate that ET-HARQ offers significant decoding complexity reductions and throughput improvements without compromising PER reliability. Moreover, to validate our findings, we have proposed a practical early termination mechanism employing belief propagation and neural network techniques, achieving prediction accuracy of up to 75% for packets encoded with practical short block length codes, i.e., BCH and CRC-polar codes. Simulation results confirmed that the BP-NN-based ET-HARQ performed similarly to the



developed analysis for PER, and complexity underscoring its potential for URLLC applications.

#### APPENDIX

In a standard order- $s$  OSD decoder, the observed sequence at the receiver is first sorted to select the  $k$  most reliable bit. This requires a specific permutation order denoted as  $\kappa_1(\cdot)$  to arrange the received symbols in order of descending magnitudes of their log-likelihood ratios and applying the same permutation to the columns of the generator matrix  $\tilde{\mathbf{G}}^{(m)} = \kappa_1(\mathbf{G}^{(m)})$ . Then the Gaussian elimination is applied to obtain the systematic form of the generator matrix as  $\tilde{\mathbf{G}}^{(m)} = [\mathbf{I}_k, \tilde{\mathbf{P}}^{(m)}]$ , where  $\mathbf{I}_k$  and  $\tilde{\mathbf{P}}^{(m)}$  denotes the identity matrix and the parity sub-matrix corresponding to the generator matrix in the  $m$ -th transmission respectively.<sup>2</sup> Associated with decoding order  $s$ , the OSD decoder generates a list  $|\mathcal{L}_{\text{TEP}}(s)|$  of test error pattern (TEP) of length  $k$ , where

$$|\mathcal{L}_{\text{TEP}}| = \sum_{i=0}^{\lfloor s \rfloor} \binom{k}{i} + \left\lfloor (s - \lfloor s \rfloor) \binom{k}{\lfloor s \rfloor + 1} \right\rfloor. \quad (26)$$

where the order  $s$  is allowed to be non-integer. This TEP list comprises all  $k$  length sequences with hamming weight less than  $\lfloor s \rfloor$  and  $\left\lfloor (s - \lfloor s \rfloor) \binom{k}{\lfloor s \rfloor + 1} \right\rfloor$  most probable TEPs with Hamming weight  $\lfloor s \rfloor + 1$ . Finally, let  $\mathbf{e}_i$  denote the  $i$ -th TEP in the list. The set of candidate codewords is computed by flipping the hard-decoded received sequence  $\mathbf{r}$  according to each TEP and mapping as

$$\tilde{\mathbf{c}}_i = (\mathbf{r} \oplus \mathbf{e}_i) \tilde{\mathbf{G}}^{(m)} \quad (27)$$

Therefore, the complexity of OSD is dominated by the total number of TEPs. Then, the total number of binary operations at a given packet length  $n_m$  at the  $m$ -th transmission round is given as

$$K_f^m(s) = n_m k + \frac{|\mathcal{L}_{\text{TEP}}|}{2} \left( n_m - q + \frac{q n_m}{k} \right) \quad (28)$$

where  $q$  is the number of quantization bits. In (28), the complexity is considered as the worst-case complexity, which is fixed with respect to the decoding order  $s$ .

There are many improvements in the standard OSD decoding that can reduce decoding complexity when the received codeword has higher reliability. For example, when the received block is decodable, OSD can apply a “stopping criterion” to early terminate the decoding process once a  $\tilde{\mathbf{c}}_i$  is likely to be the correct decoding outcome. Effective stopping criteria were investigated in the segmentation-discarding method [10] and PB-OSD [37].

We obtain the complexity  $K_s^m$  with PB-OSD via the simulation. Specifically, for a given SNR, an order- $s$  PB-OSD is tuned to achieve the near ML decoding according to [37]. Then, the average number of TEPs of such a PB-OSD for only decoding decodable blocks is measured from the simulation. Denoted by  $|\mathcal{L}'_{\text{TEP}}|$  this average number of TEPs,  $K_s^m$  is obtained similarly to (28), i.e.,

$$K_s^m(s) = n_m k + \frac{|\mathcal{L}'_{\text{TEP}}|}{2} \left( n_m - q + \frac{q n_m}{k} \right). \quad (29)$$

<sup>2</sup> $\tilde{\mathbf{G}}^{(m)}$  may require another minor permutation to make its columns linearly independent. We ignore these extra terms for the sake of simplicity.

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